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Research article

Influence of SunLight on the Cyanobacterial Diversity of Rice Fields.

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ABSTRACT

In the tropical paddy fields, numerous prevailing conditions and factors i.e., either natural or man-made, apparently influence the cyanobacterial diversity. On the basis of study on different rice cultivation sites around the Varanasi district of Uttar Pradesh (India), it had been observed that the gradually increasing dense canopy of rice plants resulted in the death of already flourishing filamentous green algal mat and favored the growth of various cyanobacterial forms. In most of the rice fields, the maximum diversity of cyanobacteria was during the vegetative growth phase and grain filling phase of rice cultivation, when maximum shade had been provided by the rice plants. In most of the rice fields around the last phase of rice cultivation, when the soil became fully exposed to solar light, the brownish coloured papery layer of Aulosira sp. had been observed on the soil as well as attached to the remaining stumps of the rice plants but the cyanobacteria was in the sporulation phase. However, an appreciable number of Nostoc, Cylindrospermum, Microcoleus and Chroococcus sp. had been observed from the fully exposed and illuminated portion of rice fields and its cultivation cycle. Most of the cyanobacterial forms collected from the fully exposed portions of rice fields to sunlight, developed larger amounts of bright/ dark coloured mucilage or exopolysaccharide compared to most of the other forms that grew in the shelter of paddy. The present communication is related to the study of local cyanobacterial diversity along with its interaction with the light regime that can be useful for the efficient algalization.

Keywords: Cyanobacteria, Rice fields, Sunlight

Type of article: Laboratory research

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INTRODUCTION

Cyanobacteria is a diverse taxonomic group of prokaryotes that have similarities with both the gram-negative bacteria as well as the eukaryotic algae (Cruz et.al., 2020, Kumar et al., 2018). As a photoautotrophic organism, it contains chlorophyll-a along with other light harvesting pigments especially the phycobilin i.e., blue phycocyanins and allophycocyanin as well as red phycoerythrin (Luinstra, 2018). Therefore, as a result of its special pigmentation, cyanobacteria can utilize orange, yellow and green portions of visible spectrum (5000 to 6500 Å) which is rarely utilized by other photoautotrophs and it maintain relatively high growth rate even at low light intensity (Luuc et al., 1999). The cyanobacteria are known to respond towards light intensity and quality, by changing its pigmentation (Tandeau et al., 1993). Quesada et al., 1998 said that "The adaptation to low light could be related to the presence of phycoerythrin, which expands the range of wavelength absorption". Cyanobacteria require moisture and diffused light for their growth, so ubiquitous in distribution, however the tropical rice fields are the most popular habitat for them.

Rice fields as an agronomically managed temporary wetland ecosystem sustains tremendous cyanobacterial diversity (Bambaradeniya, 2003). In paddy fields, there is a remarkable succession of both planktonic as well as terrestrial cyanobacterial forms throughout its cultivation cycle. The different prevailing conditions and factors of paddy fields i.e., either man made or natural, obviously influence its cyanobacterial diversity (Sharma, 2018). Increased leaf area index of rice plant resulted in shading of under canopy, so light was an important factor for the species' succession in rice fields (Zevenboom and Mur, 1980).

The understanding of local cyanobacterial population and their interaction with various natural factors as well as agricultural practices, might increase the efficiency of algalization (Sharma, 2018). However, a lot of work had already been done

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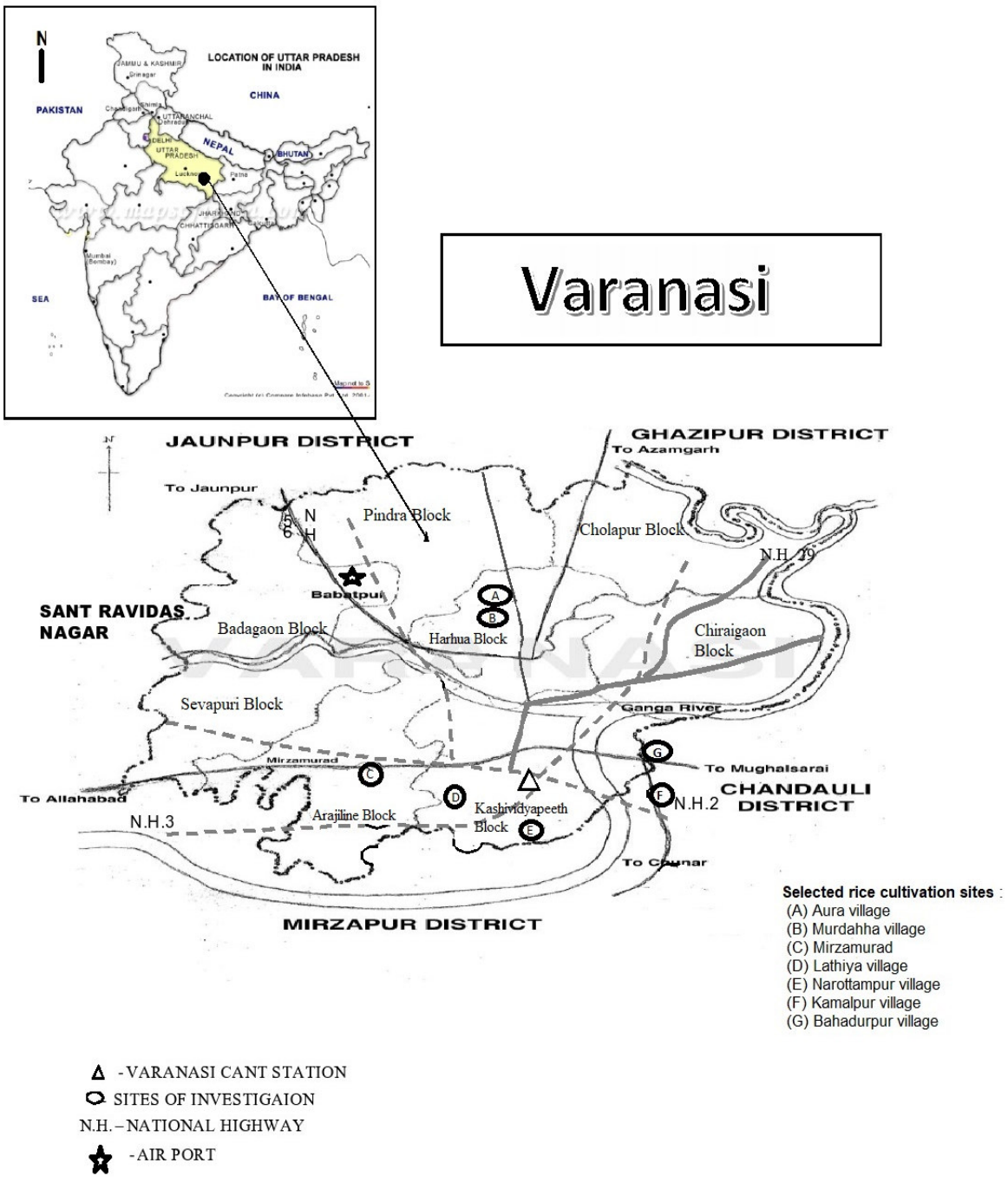


Figure 1. Diagrammatic map of Varanasi district showing the location of investigated rice cultivation sites

on the cyanobacteria but most of which were only confined to in vitro studies. Keeping the above-said facts in view the present communication pertains to an interpretation of in vivo effect of light regime on rice field cyanobacterial diversity.

MATERIALS AND METHODS

Varanasi lies at 25° 18' N latitude and 83°1' E longitude in the eastern part of the middle Indo-Gangetic plain (IGP). It is situated about 76.19 meter above the sea level and drained by the river Ganges and its tributaries. The climate of Varanasi and its surrounding areas are typical of the middle IGP. The year enjoys all the three important seasons, i.e., rainy, winter and summer with an annual rainfall of about 1,100 mm. The rainy season starts with the onset of heavy monsoon towards the end of June and continues till September, ending up with 90% annual rainfall within these months. The average sunshine hours during the rice cultivation periods vary between 3.58 to 8.57 (Table-1).

Varanasi is one of the chief rice producing districts of Uttar Pradesh, India. The soil type of Varanasi and its surrounding

Table 1. The Meteorological conditions at Varanasi during 2000 - 2001**

Year	Month	Rain Fall Total(mm)	Temperature Average(°c)		R.H. (%)	Average sunshine hour
			Minimum	Maximum		
2000	June*	184.1	25.34	33.99	82.35	5.30
	July	292.0	25.05	33.36	94	4.25
	August	05.40	25.38	33.44	84.38	6.39
	September	396.4	22.77	31.46	89.8	4.58
	October	000.0	19.03	33.74	81.06	8.57
	November	000.0	13.34	30.21	81.9	7.28
	December	000.0	7.54	24.75	83.54	7.96
2001	June*	183.2	25.82	34.96	78.23	4.40
	July	380.5	25.64	32.86	87.25	3.58
	August	294.2	25.83	33.52	88.83	5.44
	September	288	24.28	33.34	86.46	5.99
	October	158.6	21.54	31.95	87.06	3.85
	November	000.6	15.46	29.78	88.23	7.77
	December	000.0	10.03	22.96	92.54	5.73

* From 15th June and onwards
** Data obtained from I.M.D., Department of Geophysics, B.H.U.

areas varies from various mixtures of sand and clay (Table-2). In this area, the cultivation of paddy begins in the month of June last or July and extends mostly up to the month of November (Table-2).

Table 2.

Soil Property	Method Used	Reference
Soil Texture	Hydrometric method	Baruah and Barthakur, 1997
Soil organic carbon	Chromic acid digestion and gravimetric determination of CO_2	Walkley and Black, 1934
Soil available phosphorus	Olsen's method	Olsen *et al.,* 1954

The collection of cyanobacterial/ soil samples from seven selected rice cultivation sites i.e., (A) Aura village; (B) Murdahha village; (C) Mirzamurad; (D) Lathiya village; (E) Narottampur village; (F) Kamalpur village; (G) Bahadurpur, continued for two crop seasons i.e., during 2000 and 2001. These paddy fields were situated around and near Varanasi district of Uttar Pradesh (India) (Fig-1; Table-2).

The rice fields were examined during the five phenophases of cultivation i.e., 1) pre-plantation phase (1st phase); 2) transplantation phase (2nd phase) i.e., up to 20 days after the transplantation of rice plants; 3) vegetative growth phase (3rd phase) i.e., between 21 to 60 days after rice transplantation; 4) grain filling phase (4th phase) i.e., between 61 days up to the harvest; 5) post-harvest phase (5th phase) i.e., up to 10 days after the harvest of rice crop. The soil samples were collected separately both from the middle as well as the edge of the field.

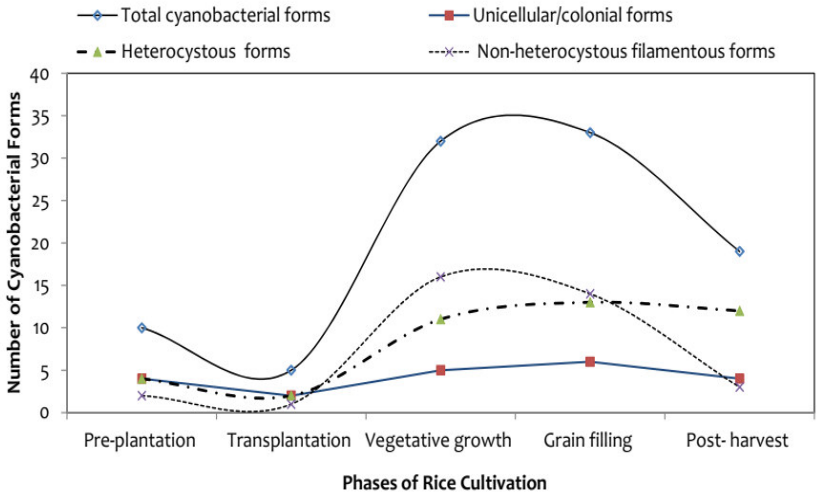


Figure 2. Cyanobacterial succession in different phases of rice cultivation. Actually, observation of transplantation and vegetative phase was recorded up to 20 days and 60 days after transplantation, respectively.

Cyanobacterial culture and isolation were performed as per the procedure mentioned by Allen and Arnon (1955) modified by Arnon et al. (1974) media. All the specimens were critically examined by using keys given by Desikachary (1959) and Anand (1989).

The relative abundance of a particular cyanobacterial type was calculated by employing the formula:

Table 3. Cyanobacterial diversity in different phases of rice cultivation cycle:

S. No.	Unicellular/ colonial forms	Phases of the rice cultivation cycle					Relative Abundance (%)
		1st	2nd	3rd	4th	5th	
1	***Gloeocapsa*** *stegophila* var. *crassa*	+		+	+	+	3.88
2	***Chroococcus*** *montanus*	+	+	+	+	+	4.44
3	*C. pallidus**	+				+	2.77
4	***Aphanocapsa*** *biformis***			+	+		2.22
5	*A. banaresensis*				+	+	1.66
6	***Aphanothece*** *pallida*	+	+	+	+		2.22
7	*A. bullosa***			+	+		1.11

S. No.	Non-heterocystous filamentous forms	Phases of the rice cultivation cycle					Relative Abundance (%)
		1st	2nd	3rd	4th	5th	
1	***Microcoleus*** *chthonoplastes*	+	+	+	+	+	5.50
2	***Spirulina*** *laxissima* **			+			0.55
3	***Phormidium*** *ambiguum*	+			+	+	2.77
4	*P. corium***				+		0.55
5	*P. anomala* **			+	+		1.66
6	***Lyngbya*** *dendrobia*				+	+	1.66
7	*L. .aestuarii* **			+	+		2.22
8	*L. major* **				+		1.11
9	*L. nordgardhii* **				+		0.55
10	*L. rubida* **			+			1.11
11	*L. martensiana* var. *minor***			+			1.11
12	*L. kashyapii* **			+			1.11
13	***Oscillatoria*** *tenuis***			+			1.11
14	*O. subbrevis* **			+			0.55
15	*O. willei***			+	+		1.66
16	*O. laete-virens* **			+			0.55
17	*O. animalis***			+	+		2.22
18	*O. terebriformis***				+		1.11
19	*O. foreaui* **			+	+		2.22
20	*O. perornata* f. *attenuata* **			+			0.55
21	*O. fomosa***			+	+		2.22
22	*O. claricentrosa* f. *bigranulata***			+	+		2.77

S. No.	Heterocystous forms	Phases of the rice cultivation cycle					Relative Abundance (%)
		1st	2nd	3rd	4th	5th	
1	***Cylindrospermum*** *licheniforme*				+	+	2.77
2	*C. muscicola*			+	+	+	4.44
3	*C. gorakhpurense*			+		+	2.77
4	***Anabaena*** *fertilissima*	+	+	+			2.22
5	*A. iyengarii***			+	+		2.22
6	***Nostoc*** *spongiaeforme* var. *varians*	+			+	+	4.44
7	*N. linckia* var. *arvense*	+		+		+	3.88
8	*N. commune*	+	+	+			3.33
9	*N. punctiforme*				+	+	2.77
10	***Aulosira*** *fertilissima*			+	+	+	13.88
11	*A. fertilissima* var. *tenuis*			+	+	+	8.33
12	***Scytonema*** *bohneri*			+	+	+	2.77
13	*S. pseudohofmanni* **			+	+	+	3.33
14	*S. ocellatum* f. *minor**					+	1.11
15	*S. guyanense* var. *prolifica* **				+	+	2.22
16	***Calothrix*** *braunii***			+	+		2.22
17	*C. marchica* var. *intermedia***				+		1.66
18	***Gloeotrichia*** *natans* **				+		1.11

(*-obtained only from exposed portion; **- obtained only from unexposed portion)

$$Relative\ abundance\ (RA) = \frac{Y}{X}\ x\ 100$$

Where, X = Total numbers of samples collected; and, Y = Number of samples from which a particular cyanobacterial type was identified.

The physico-chemical properties of air-dried soil samples from rice fields were analysed as per procedure mentioned hereunder (Table-2).

RESULTS

Observation from the rice field at different locations clearly reveals that in total forty-seven cyanobacterial forms were obtained. The highest diversity was shown by the non-heterocystous filamentous with twenty-two forms followed by heterocystous filamentous with eighteen and unicellular/ colonial with seven forms (Table-3). Further, among the different phenophases, the maximum diversity of cyanobacteria was observed during the grain filling phase (33 forms) followed by vegetative growth phase (32 forms) of rice cultivation, when the maximum shade had been provided by the crop canopy. However, only nineteen forms were obtained from the post harvested condition of rice fields. Out of all phases, the second phenophase recorded the minimum number (5) of cyanobacterial forms (Fig 2; Table 3).

The highest number of cyanobacterial populations (57.45 %, 27 forms) were obtained from unexposed portions of rice fields, i.e., under rice canopy, while only two (4.25%) forms from fully exposed portions. However, eighteen (38.30 %) forms recorded from both the exposed as well as unexposed portions of rice fields (Fig.3 & 4). Furthermore, about 61.2 per cent (11 forms) of heterocystous forms were obtained from both the exposed as well as unexposed portions of rice fields, while most of the non heterocystous filamentous forms (86.36 % i.e., 19 forms) were observed from the unexposed conditions (Fig 4; Table 3).

An appreciable growth and comparatively higher relative abundance of Nostoc, Cylindrospermum, Microcoleus and Chroococcus sp. had been observed from the fully exposed and illuminated portions of rice fields. These forms showed a larger amount of bright/ dark or variously coloured exopolysaccharide, compared to the collections away from direct sunlight exposure (Fig. 6 a & c).

Few cyanobacterial forms, like, Microcoleus chthonoplastes and Chroococcus montanus had been observed from all the five phenophases of rice cultivation (Table 3). Out of which, Microcoleus sp with higher relative abundance was looking to be the most tolerant to the light exposure. However, the samples having Gloeocapsa stegophila and many species of Oscillatoria collected were recorded under the crop canopy (shady areas).

Most of the unicellular/colonial forms that had been collected from the exposed portion of fields, found to be entangled between the filaments of heterocystous forms. However, the heterocystous filaments of Calothrix marchica were obtained from the field B with in the large bluish green coloured mucilaginous ball of an unicellular/ colonial form i.e., Aphanothece pallida, under the rice canopy during early grain filling phase. However, the

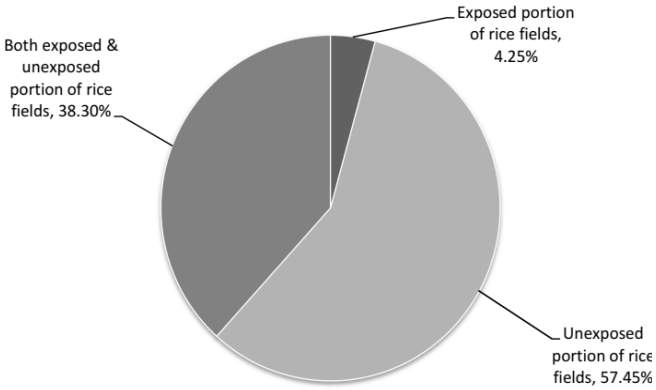


Figure 3. Cynaobacterial population in relation to different light exposure conditions.

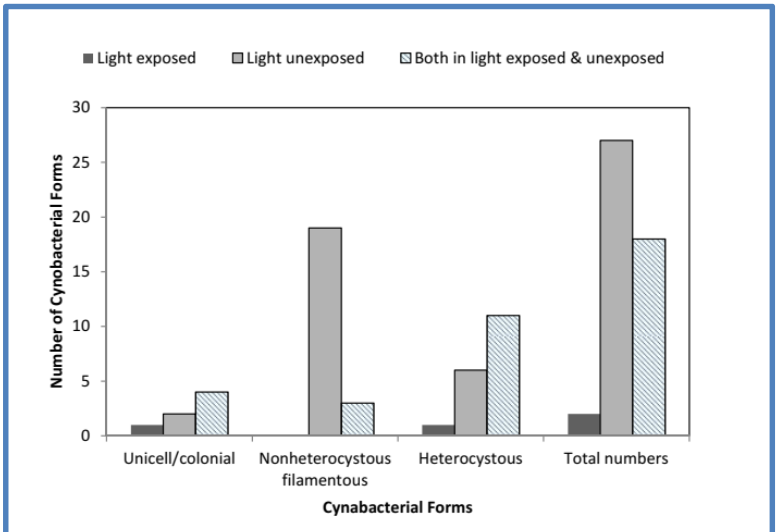


Figure 4. Cyanobacterial diversity in relation to different light exposure conditions.

unicellular/ colonial forms were not showing any clear-cut pattern of diversity related to the light regime (Fig. 2).

Visual observation from the field shows that the cyanobacterial forms collected from somewhat deeper layers of soil profile were not fully active, mostly they are in the form of spores. Further, transplanted rice crops with closer row spacing (fields A, B, C and G) produce higher cyanobacterial diversity and thick mat formation as compared to wider row-spacing fields (fields D, E and F).

In most of the fields (Fields A, B. C, F), around the last phenophase of rice cultivation, when the rice field soil became fully exposed to sunlight, brownish coloured prominent thick papery layer of Aulosira had been seen on the soil surface as well as attached to the remaining stumps of the paddy plants. The cyanobacterial layer was easily removable, having its sporulated filaments (Fig.5-b).

DISCUSSION

A lot of prevailing conditions of the paddy fields particularly, water logging, high temperature, neutral to alkaline pH, light intensity and nutrients availability etc. generally encourage the luxuriant cyanobacterial growth (Amarawansa, et al. 2018; Kondo and Yasuda, 2003; Sharma et al. 2014). Amongst the major physical conditions, the light regime is the most important one, influencing the seasonal fluctuation of paddy field's cyanobacterial diversity. (Quesada et al., 1998).

Those flooded paddy fields reflected high cyanobacterial growth that had slow movement of irrigation water along with low turbidity (Bunt, 1961). Least appearance of cyanobacteria at the 2nd (transplantation) phase during rainy days was due to frequent churning of soil during the puddling process (Roger and Kulasooriya, 1980; Sharma et al., 2014). During that period, the cyanobacterial forms appeared only at the side of rice fields, particularly few days after the rain showers, probably because being a photoautotroph it required at least some sunlight to perform photosynthetic process. On the other hand, well developed visible growth of Microcoleus chthonoplastes, along with Nostoc commune was observed even in the second phase of field 'E' (Narottampur village), where there was no proper water logging due to poor irrigation facility and high percentage of sand in the soil (Table-2) (Sharma et al., 2014).

With the maturation of rice crop there was gradually increase in shading, thus, light became an important factor determining the cyanobacterial species succession in rice fields (Zevenboom and Mur, 1980). Cyanobacteria are probably better suited photoautotrophs for low-light conditions created by the rice plants which resulted in the death of thick floating layer of filamentous green algae, that encouraged its development during the vegetative growth phase of rice cultivation (Luuc et al., 1999; Oh and Rhee, 1991; Sharma and Singh, 2011). A month after the transplantation of rice, the amount of light reaching the underlying water surface might decline to as much as 85% and after the second month by 95% (Halwart and Gupta, 2004; Roger, 1996). As a result, in the present case rice canopy appeared to cause an attenuation of solar light reaching the lower side during the 3rd and 4th phenophase of cultivation (Sharma and Singh, 2006). In fact, in the early sixties, Singh (1961) also reported the beneficial effect of the plant canopy shading on the underlying cyanobacteria, in the fields of maize, sugarcane, as well as in the grasslands.

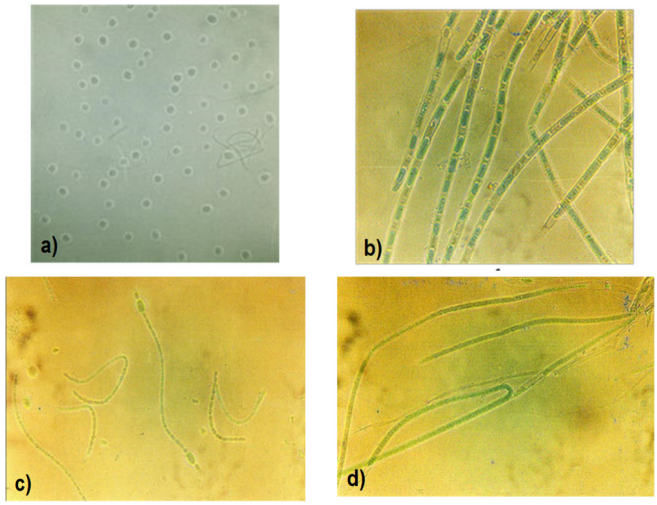


Figure 5. Microscopic photographs of various cyanobacterial forms: **a)** Apahanocapsa banarsensis, **b)** Aulosira fertilissima Ghose var. tenuis Rao, in stage of sporulation, **c)** Cylindrospermum gorakhpurense and **d)** Scytonema bohneri Schmidle.

The cyanobacterial diversity was found to be very less during the exposed period i.e., 1st, 2nd and 5th phenophase of the rice cultivation cycle in comparison to the 3rd and 4th phase having good crop canopy (Fig. 2, 3 and 4; Table-3). Most of the cyanobacterial forms are generally sensitive to intense solar light (Luuc et al., 1999; Reynaud and Roger, 1978) and could be regarded as low light species (Brown and Richardson, 1968), such as the samples having Gloeocapsa stegophila and many species of Oscillatoria were mostly collected from shady areas, corners of the fields away from direct sunlight (Fremy, 1925; Luuc et al., 1999). In the present investigation, the most dominant cyanobacteria i.e., Aulosira



Figure 6. **a)** Photograph of newly developing light coloured floating colonies of cyanobacteria in a flooded field, along with some brownish coloured decaying filamentous green algal mass beneath the rice plants, **b)** Blue green coloured scattered patches of *Cylindrospermum* sp., floating on the paddy field water and **c)** Multicoloured mucilaginous layer of cyanobacteria developed on the outer portion of paddy field water, that was fully exposed to direct solar radiation

fertilissima and A. fertilissima var. tenuis, appeared during the 3rd, 4th and 5th phenophase of rice cultivation (Table 3), which coincide with the month of September and October, when rice plants made a dense canopy. The Aulosira sp. came up gradually in vegetative state and formed the thick yellowish blue-green to light brown gelatinous layer (Sharma, 2006). At the end of rice cultivation, when the soil became fully exposed to sunlight, brownish coloured dry papery layer of sporulated Aulosira sp. had been observed on the soil as well as attached to the remaining stumps of the rice plants (Roger and Kulasooriya, 1980; Singh, 1942 & 1961; Singh, 1976; Sharma and Singh, 2011). Generally, as a result of fast growth of Aulosira sp. in tropical paddy fields, its prominent multiple successional layers had been seen one above the other (Roger and Kulasooriya, 1980; Singh 1961).

The another heterocystous cyanobacterium i.e., Cylindrospermum sp. was reported to grow on the side of rice fields, on the surface of open and calm portions of the flooded fields 'A' and 'C', from the side of field 'B' on the moist soils not shaded by rice plants and after the harvesting phase in fields 'B' and 'C'. All these conditions show direct positive relation between Cylindrospermum sp. and sunlight (Fig. 5-c & 6-b) (Roger and Kulasooriya, 1980; Traore et al., 1978). Large amount of Nostoc sp. had been collected from the early phases of the rice cultivation cycle and also after the harvest of rice crop (Table-3), when the field was fully exposed to solar irradiation (Quesada et al., 1998). Dodds et al., (1995) reported the high adaptability of Nostoc sp towards a wide range of light conditions.

The cyanobacterial forms that had been collected from portions having direct exposure to sun light, showed a large amount of coloured mucilage or polysaccharide sheath compared to the forms that grew on portions away from direct light, in the shelter of the rice plants (Fig. 6-a & c) (Sharma, 2006). The exopolysaccharide protects cyanobacteria in various means especially against the damaging effect of the solar irradiation (Kumar et al., 2018; Whitton, 2000) The production of variously coloured mucilage by cyanobacteria was might be due to the development of photoprotective carotenoid pigments (Paerl et al., 1983). As a result of exposure to the intense sun light, many cyanobacteria are reported to produce a yellowish brown colured ultraviolet light absorbing dynamic pigment i.e., scytonemin (Fleming and Castenholz, 2007).

The negligible number of non-heterocystous filamentous forms were reported from the exposed portions of rice fields, which was followed by the unicellular /colonial forms, while the heterocystous forms (Table- 3; Fig 2 & 4) were found to reflect their luxuriant growth in such an exposed condition. It indicates the higher tolerance level and adaptability of heterocystous forms (Sharma, 2006; Sharma and Singh, 2011). Except Microcoleus chthonoplastes and a few species of Phormidium and Lyngbya, all other non-heterocystous filamentous forms were confined to the shady portions of rice fields, away from the intense sunlight i.e., only reported from the third and fourth phase of rice cultivation (Table- 3). In contrast to it, Quesada et al. (1998) reported an abundance of non-heterocystous cyanobacteria under high light intensity in the rice fields of Valencia, Spain, that might be due to a huge difference in the climatic condition of both the areas.

Apart from the upper soil surface, many a time cyanobacteria also inhabit in the deeper in soil profile at complete darkness and later on, the tillage operations tend to bring the buried forms near the surface or vice versa. This might be one of the probable reasons for the year to year variations, in the visible cyanobacterial diversity in agricultural fields (Sharma, 2006). However, the forms collected from the somewhat deeper soil profile were not in fully active state, i.e., mainly in the form of spores (Whitton and Potts, 2000). Being a photoautotrophic organism, the majority of cyanobacteria require at least diffused light and are unable to multiply actively under complete darkness (Goyal, 1997).

Higher cyanobacterial diversity with thick mat visualized in the fields where rice plants were transplanted at narrow spacing (fields A, B, C and G), this was attributed to the fact that narrow row spacing form dense canopy and intercept most of the sunlight at the upper canopy, thus only diffuse light reach the ground, which favours the growth of cyanobacteria. In many rice fields, convincing growth of cyanobacterial forms were observed after harvesting the crop, where the farmers used to leave the harvested rows of paddy in the fields for some days. That might be due to more shade as well as availability of moisture beneath the harvested row of paddy (Sharma, 2006). Recently, on the basis of in vitro experiments, many workers claimed that the cyanobacteria can survive under very weak and filtered sunlight owing to the replacement of about 8% of chlorophyll-a (Ch-a) by chlorophyll-f (Ch-f) pigments in their photosystem I (PS-I), under such attenuated light conditions (Gisriel, et al., 2020, Nürnberg, et al., 2018, Shen, et al., 2019). Bambaradeniya (2003) highlighted that ways of rice cultivation serve as the dominant factor that controls the overall dynamics of cyanobacterial biodiversity in the rice fields.

CONCLUSION

The results obtained clearly indicated that individuals marking attendance can be potential source for contributing and transmitting their microflora through biometric devices. This indicates that a potential health risk existed at the time of study. Disinfection both before and after using the biometric machine is strongly recommended as this is able to maximize health benefits both to the individual and to the broader employee population.

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